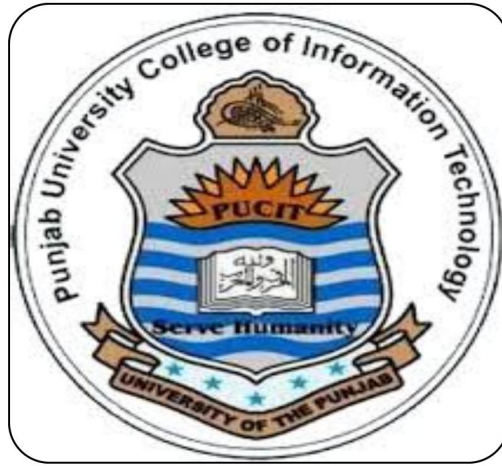


SOFTWARE ENGINEERING PROJECT DOCUMENTATION



Project Title: Student Result Management System

Subject Name: Software Engineering

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PROJECT REPORT

Problem Statement

Many educational institutes still manage student results manually or by using basic tools such as Excel sheets. This often leads to:

- Data inconsistency and duplication
- Calculation and grading errors
- Time-consuming result preparation
- Difficulty in maintaining historical records
- Lack of transparency for students
- Limited reporting and analysis for management

These limitations highlight the need for a secure, accurate, and efficient computerized system.

Proposed Solution

The proposed system is a **Java-based Student Result Management System** that provides a centralized platform where:

- **Admin** manages students, teachers, courses, and subjects
- **Teachers** enter and update student marks
- **Students** view their academic results
- **Management** generates reports and analyzes performance

The system uses a Java GUI for the frontend integrated with a database to ensure accuracy, security, and ease of use.

Project Objectives

- Automate result calculation and grade generation
- Reduce manual errors and paperwork

- Provide role-based access (Admin, Teacher, Student)
- Generate result reports efficiently
- Improve transparency and academic monitoring

Major Modules of the System

1. User Management Module

- Admin login
- Teacher login
- Student login
- Role-based access control

2.Student Management Module

- Student registration
- Student profile management (Roll No, Name, Department, Semester)
- Update and delete student records

3.Course & Subject Management Module

- Add courses and subjects
- Assign subjects to semesters
- Assign teachers to subjects

4.Marks & Result Management Module

- Enter marks (Midterm, Final, Quiz, Assignment)
- Automatic total calculation
- Grade generation (A, B, C, etc.)
- GPA and CGPA calculation

5.Result Viewing Module

- Students can view results
- Teachers can review submitted marks

6. Reports & Analysis Module

- Class-wise result reports
- Student performance analysis
- Printable result sheets

INTRODUCTION

Purpose:

The purpose of this document is to describe the requirements of the Student Result Management System. This system automates student result management.

Scope:

The system allows Admins to manage students, Teachers to enter marks, and Students to view results.

Assumptions and Dependencies

- Users have basic computer knowledge
- The system is used within an educational institute
- Required hardware and software are available

User Classes:

Admin, Teacher, Student

3. Functional Requirements**FR-1:**

Admin can add student records

FR-2:

Admin can update student records

FR-3:

Admin can delete student records

FR-4:

Login for all users

FR-5:

Role based access

FR-6:

Teacher enters marks

FR-7:

GPA/CGPA calculation

FR-8:

Student views result

FR-9:

Report generation

4. Non Functional Requirements

4.1 Usability Requirements

- The system should be user-friendly
- The interface should be easy to understand
- Screens should be simple and clearly labeled

4.2 Security Requirements

- Secure login using credentials
- Unauthorized access should be restricted
- Student data must be protected

4.3 Performance Requirements

- Fast system response
- Result calculation without delay
- Efficient handling of multiple users

4.4 Scalability Requirements

- System should handle increasing number of students
- Future enhancements should be possible

DOCUMENTATION

MAIN FLOW

- 1) Student Result Management System
- 2) User opens the Student Result Management System.
- 3) System displays the login screen.
- 4) User enters username and password.
- 5) System validates the entered credentials.
- 6) System identifies the user role (Admin, Teacher, or Student).
- 7) System redirects the user to the respective dashboard.
- 8) Admin Flow

- 8.1) User logs in as Admin.
- 8.2) System displays the Admin dashboard.
- 8.3) Admin selects the option to manage student records.
- 8.4) System displays options to add, view, update, or delete student information.
- 8.5) Admin selects the Add Student option.
- 8.6) System displays a form for student details (roll number, name, department, semester).
- 8.7) Admin enters the required information and submits the form.
 - 8.7.1. If required information is missing or invalid
 - 8.7.2. System displays an error message.
 - 8.7.3. Admin re-enters the correct information.
- 8.9) System saves the student record and displays a success confirmation message.
- 8.10) Admin selects the option to generate result reports.
- 8.11) System retrieves student results, calculates GPA and CGPA, and displays the report.

Teacher Flow

- 9) User logs in as Teacher.
- 9.1) System displays the Teacher dashboard.
- 9.2) Teacher selects the option to enter marks.
- 9.3) Teacher enters marks for quizzes, assignments, midterm, and final exams.
 - 9.3.1. If entered marks are invalid or exceed the allowed range:
 - 9.3.2. System displays an error message.
 - 9.3.3. Teacher re-enters valid marks.

9.4) System calculates total marks, assigns grades, calculates GPA, and saves the results.

Student Flow

10.1) User logs in as Student.

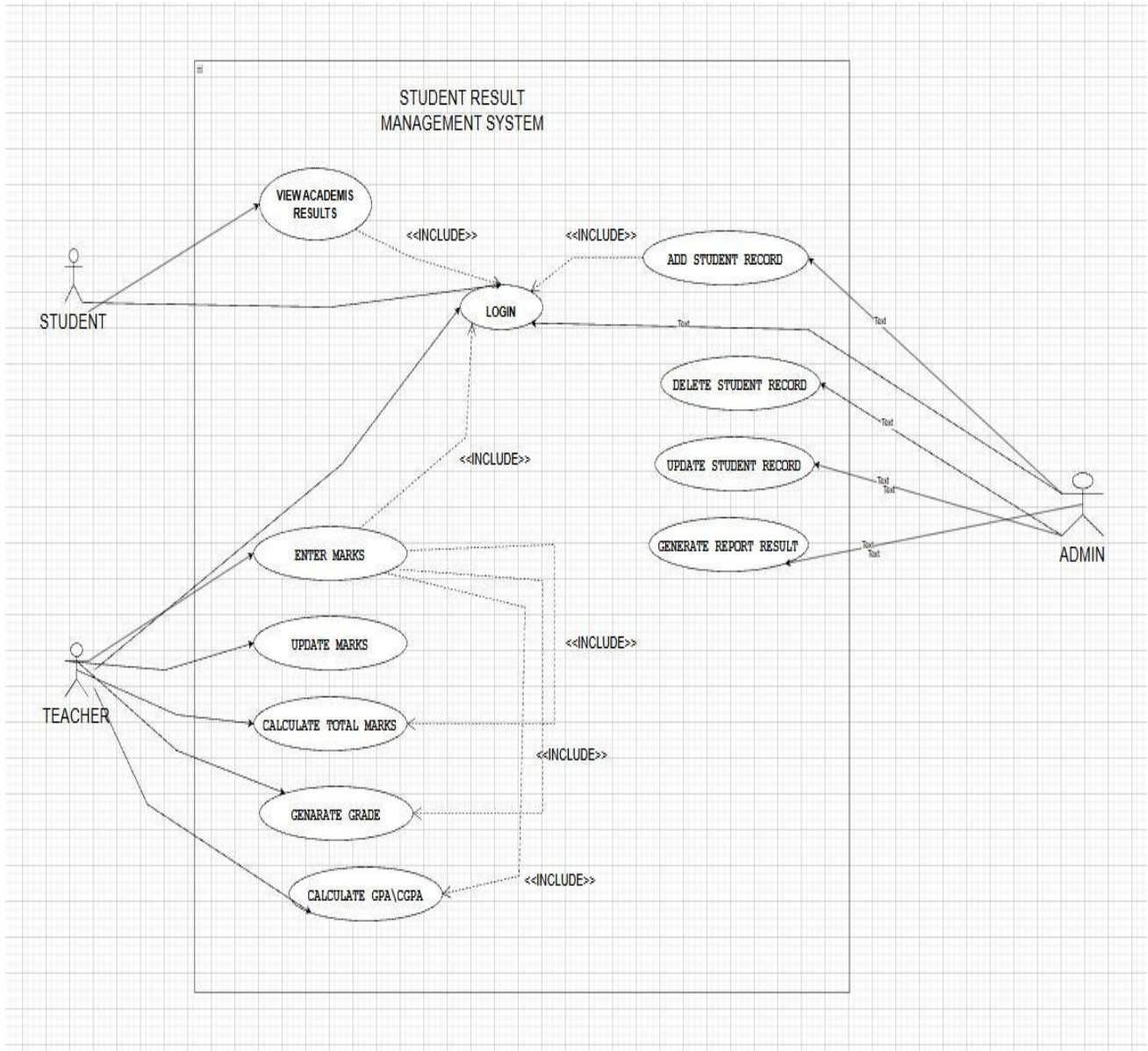
10.2) System displays the Student dashboard. 10.3) Student selects the option to view academic results.

10.3) System retrieves and displays marks, grades, GPA, and CGPA.

10.4) User logs out of the system and the session ends.

5. UML Diagram

Use Case Diagram:



This diagram shows interactions between Admin, Teacher, Student, and the system.

DOMAIN MODEL

The Student Result Management System starts when a user logs into the application using valid credentials. Based on the user's role, the system provides access to specific features.

An **Admin** can add, update, or delete student records by entering details such as *roll number, name, department, and semester*. The system stores this data securely and allows the Admin to generate academic result reports by retrieving student results and calculating GPA and CGPA.

A **Teacher** accesses the system to enter and manage student marks for *quizzes, assignments, midterms, and finals*. After marks entry, the system automatically calculates total marks, assigns grades, and computes GPA values before saving the results.

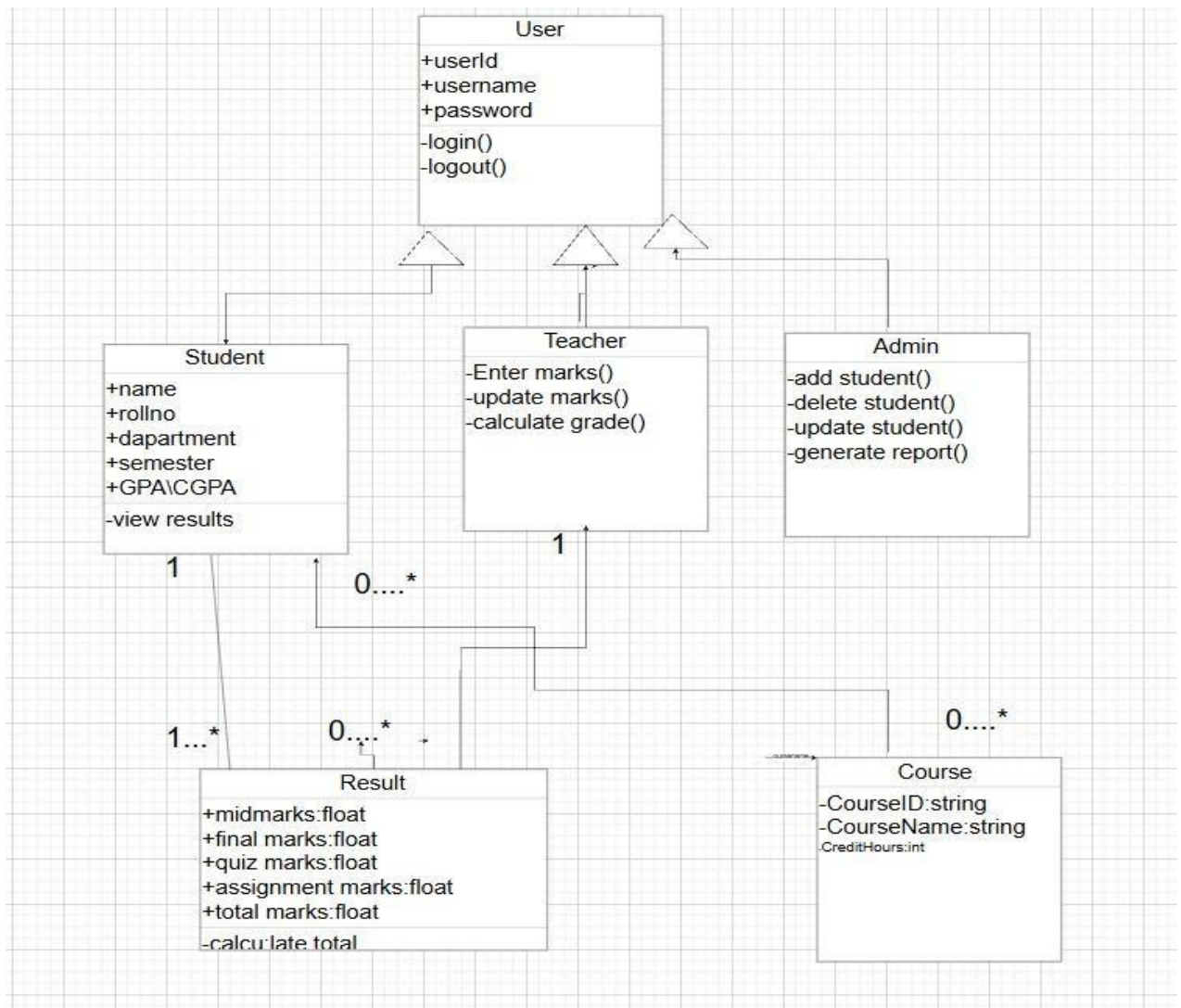
A **Student** logs in to *view* academic results only. The system displays marks, grades, GPA, and CGPA fetched from the database, without allowing any modification.

The system ensures *role-based access control* and maintains secure handling of academic data.

ENTITIES

CANDIDATES ENTITIES	DISCARDED ENTITIES
USER	SYSTEM
ADMIN	SCREEN
TEACHER	LOGIN PAGE
STUDENT	MESSAGE
COURSE	CALCULATION
RESULT	DASHBOARD
MARKS	
GRADE	
REPORT	

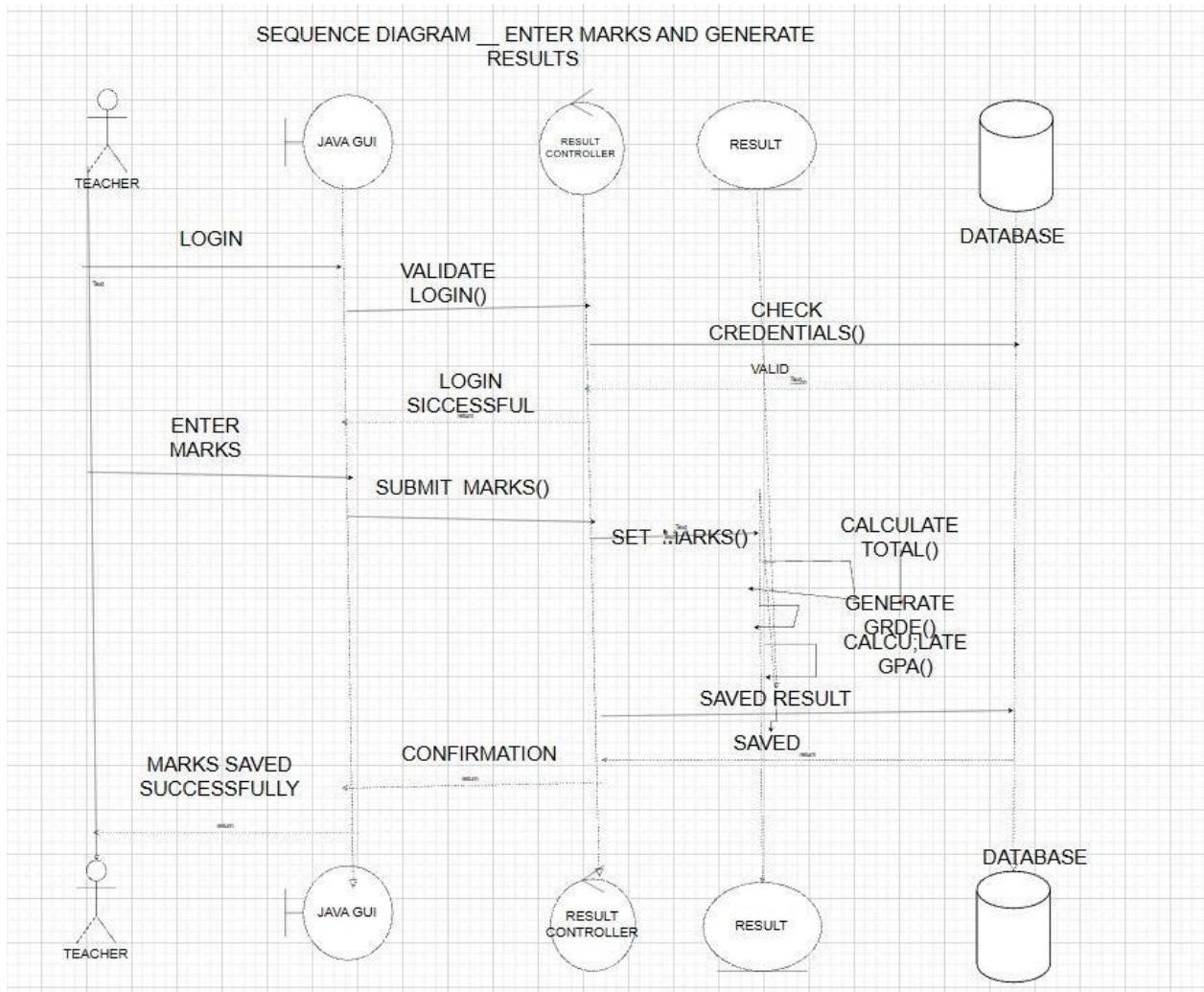
Class Diagram



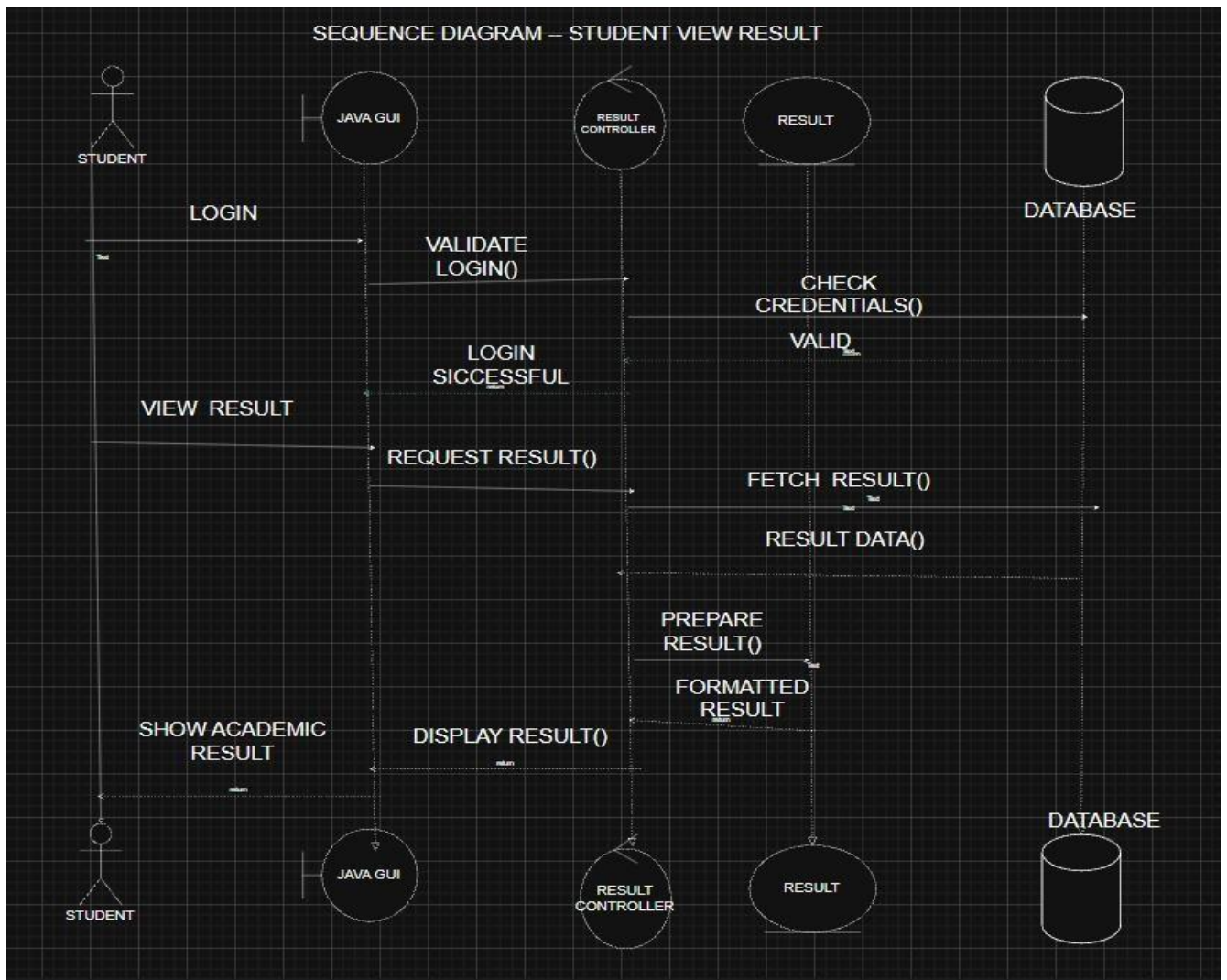
This diagram represents the system classes such as User, Student, Teacher, Admin, Result, and Report.

Sequence Diagrams

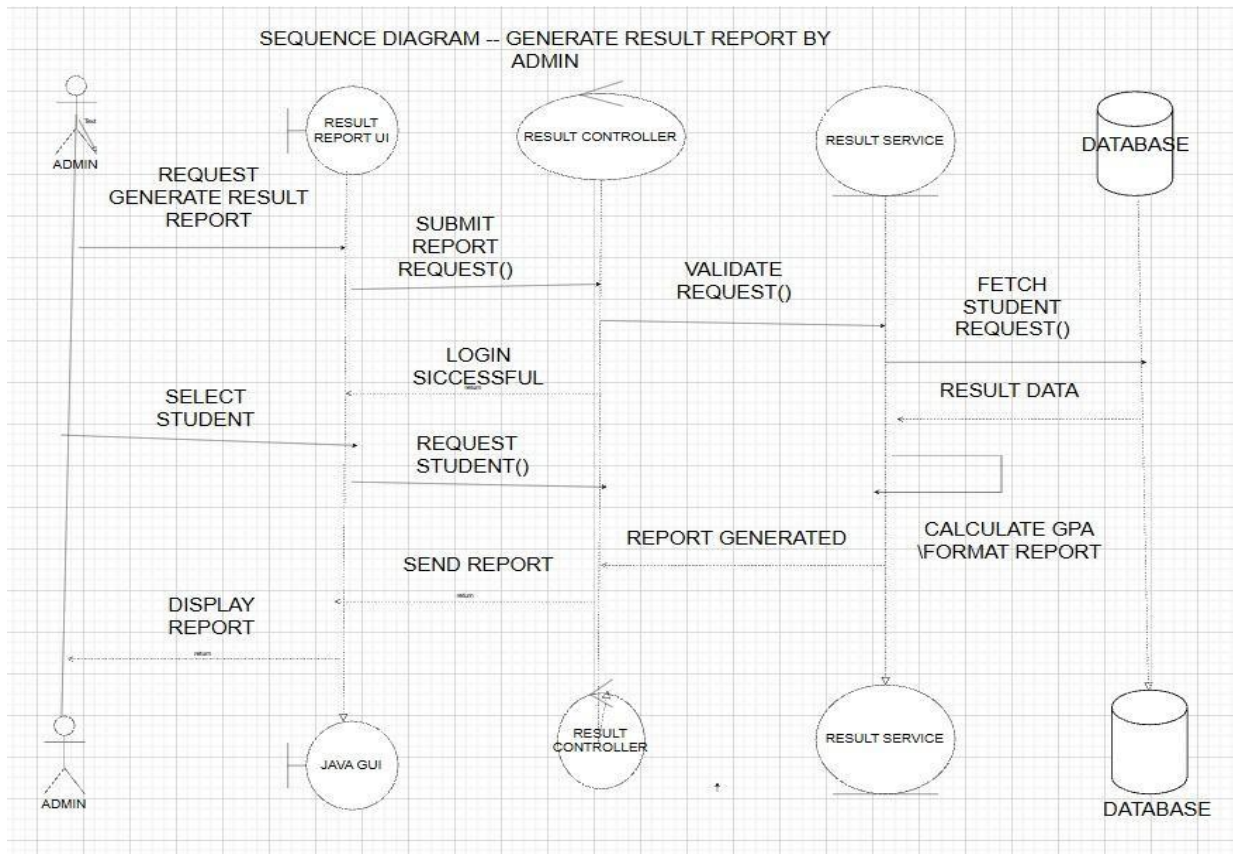
1. Teacher Enter Marks



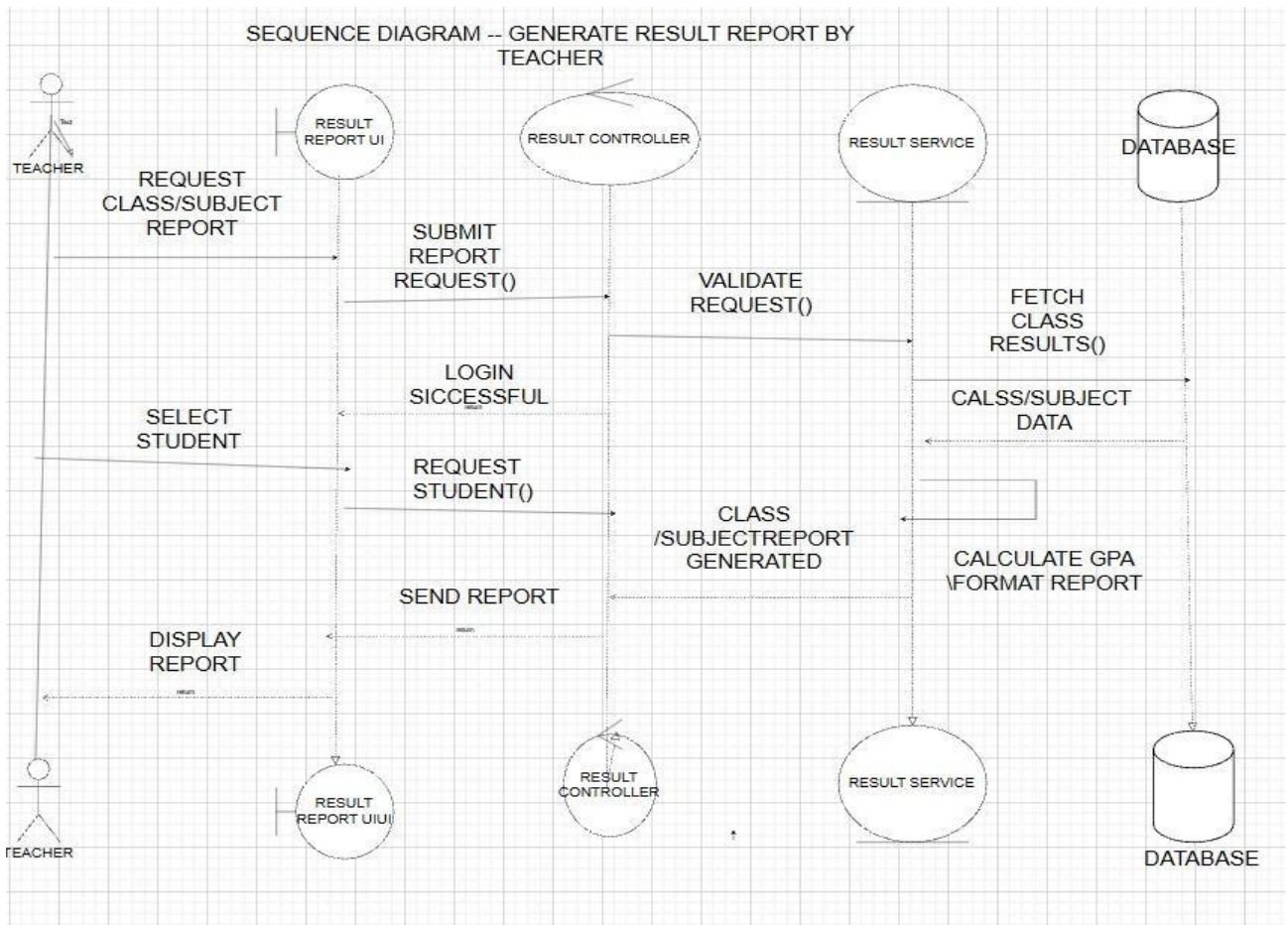
2.Student View Result



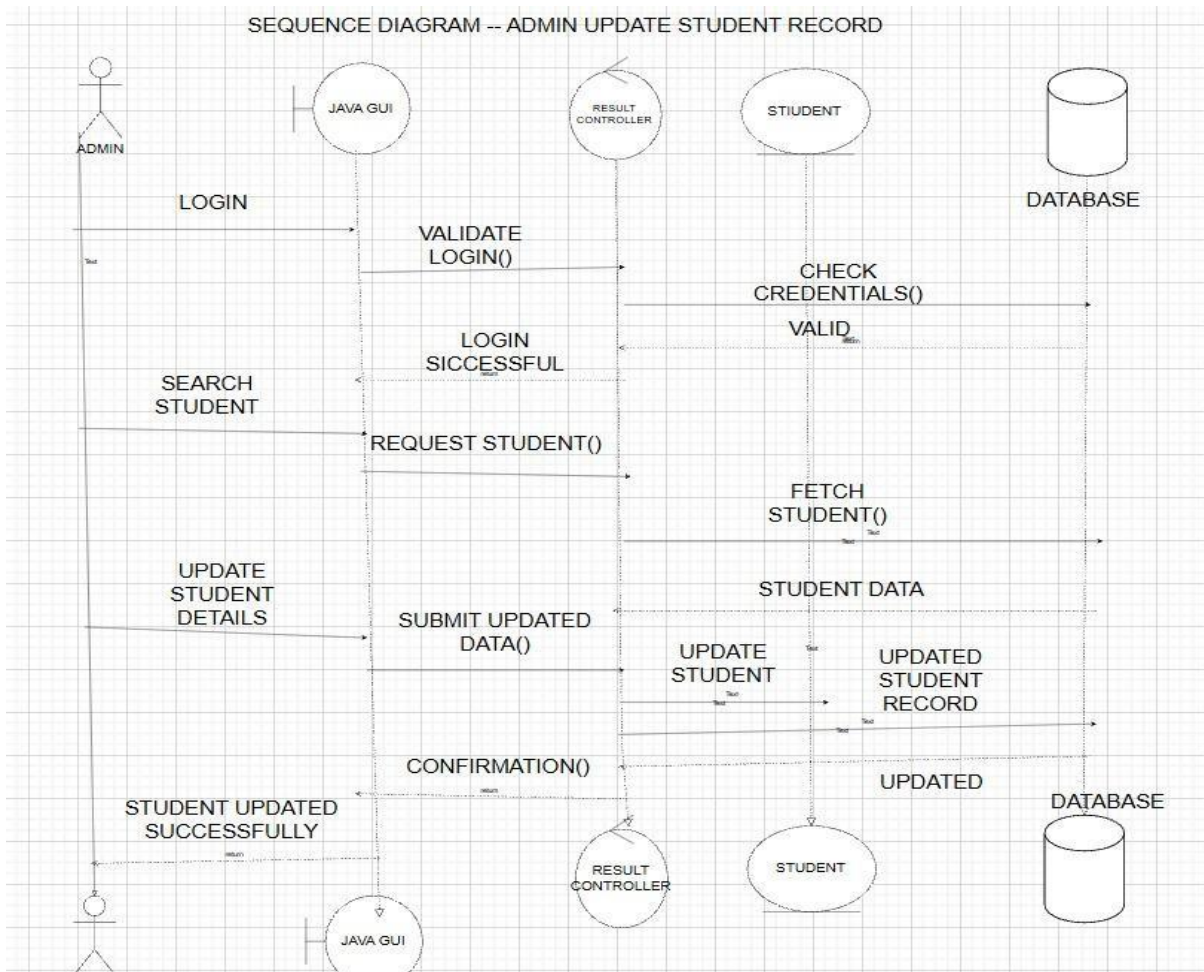
3. Generate result report by admin



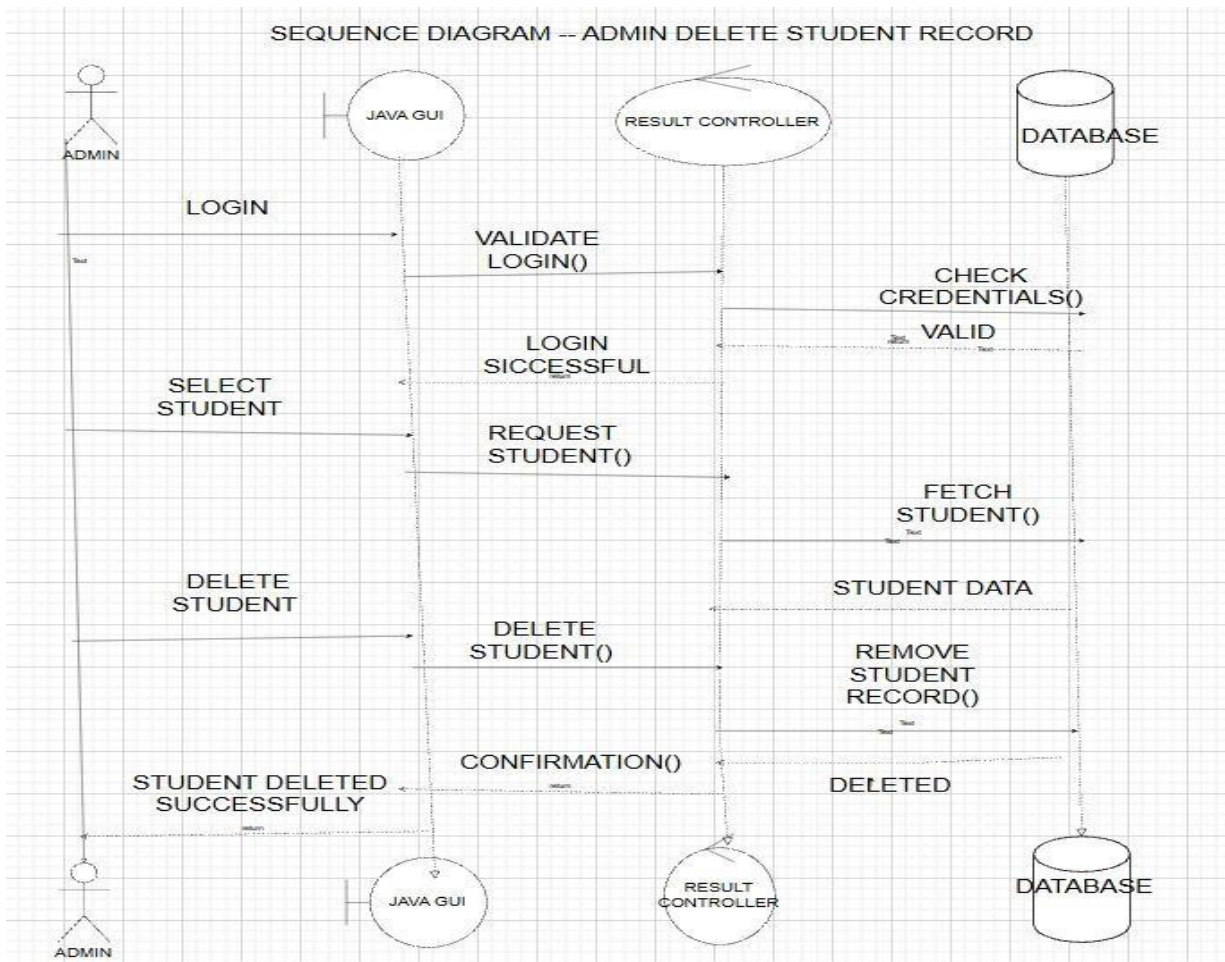
4. Generate result report by teacher



5.Admin update student diagram

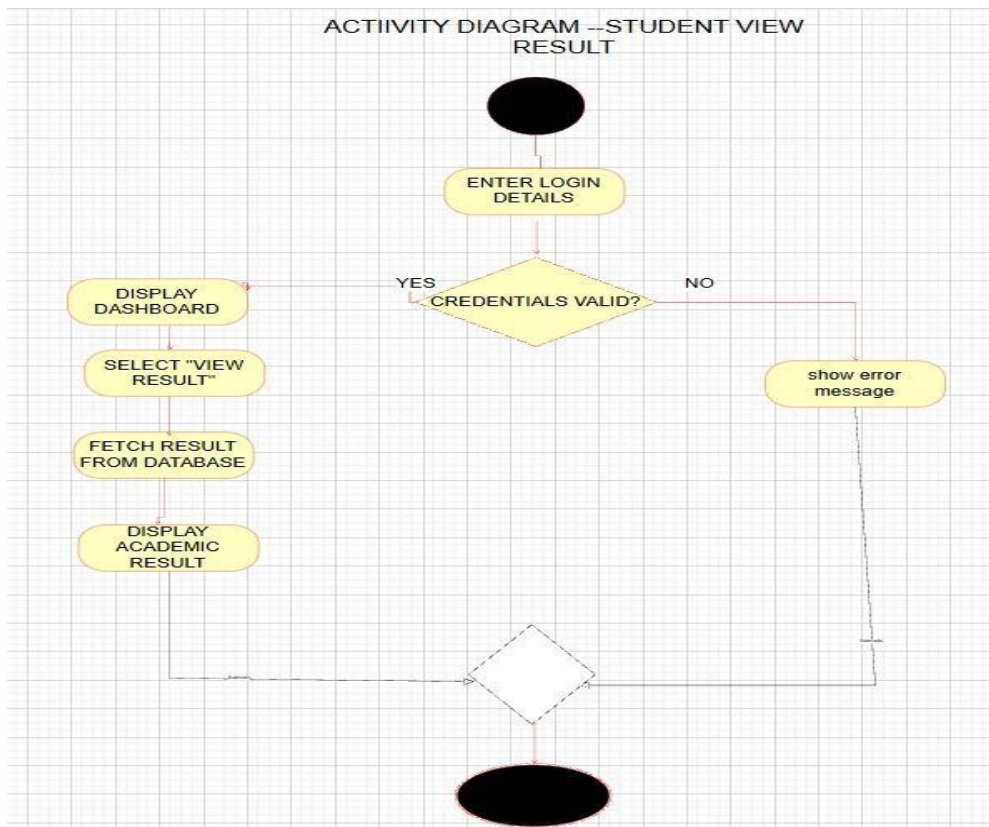


6.Admin delete student diagram

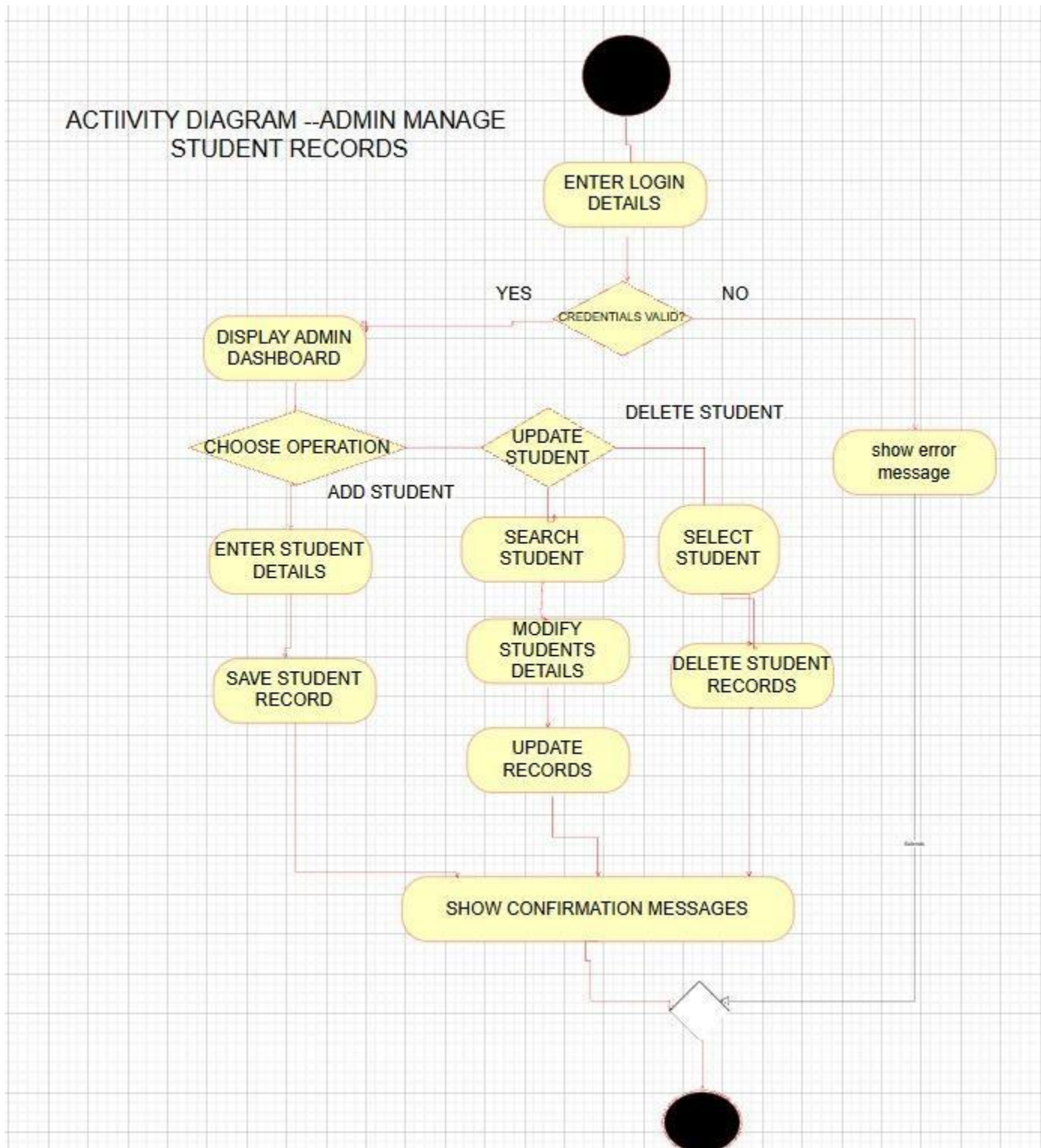


Activity Diagrams

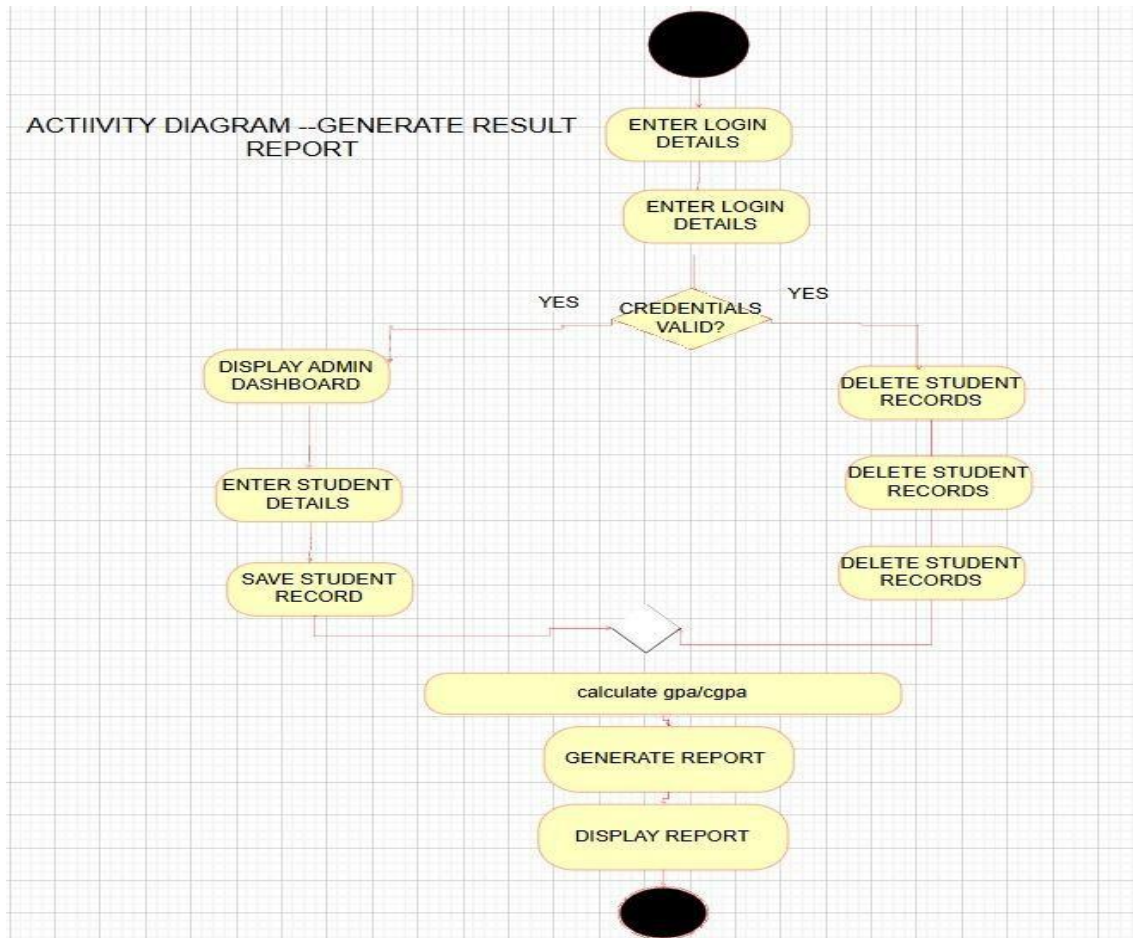
i.Student view result:



ii. Admin manage student records



iii. Generate result report



Conclusion

The Student Result Management System provides an efficient, secure, and reliable solution for managing academic results.

The system follows Software Engineering principles and uses UML diagrams for clear system modeling.